Feature Engineering & Nonlinearity

QLS–MiCM Workshop

Why This Matters

Real biological and clinical effects are **rarely linear**.

Examples:

- dose–response curves
- saturation and thresholds
- diminishing returns
- nonlinear gene/protein interactions

Key Idea: Add Nonlinear Features

We keep using a **linear model**, but expand the input features:

Polynomial Features

- [x^2], [x^3]
- interactions ([x_1x_1], [x_1^2 x_2], …)
- allows curvature and richer patterns

This is still simple, interpretable, fast.

Workflow You'll Use

1. Load the drug response dataset
2. Fit a **baseline linear model**
3. Add polynomial features (degree=3)
4. Compare model performance
5. Visualize:
    - data vs linear fit
    - data vs polynomial fit

What to Look For

- Does adding curvature improve MAE?
- Does the polynomial model better follow the data trend?
- Is the relationship between top predictor + effect nonlinear?
- When does added flexibility risk overfitting?